Unit 5. Unsupervised Learning

Learning Outcomes

- Explain the concept of clustering in data analysis and its significance.
- Compare K-means and Hierarchical Clustering and choose the appropriate technique based on the problem.
- Describe the importance of Dimensionality reduction.
- Illustrate the process of dimensionality reduction using PCA.

INTRODUCTION

- Unsupervised learning is a type of machine learning that learns from **unlabeled data**. This means that the data does not have any pre-existing labels or categories.
- The goal of unsupervised learning is to **discover patterns and relationships in the data without explicit guidance**.

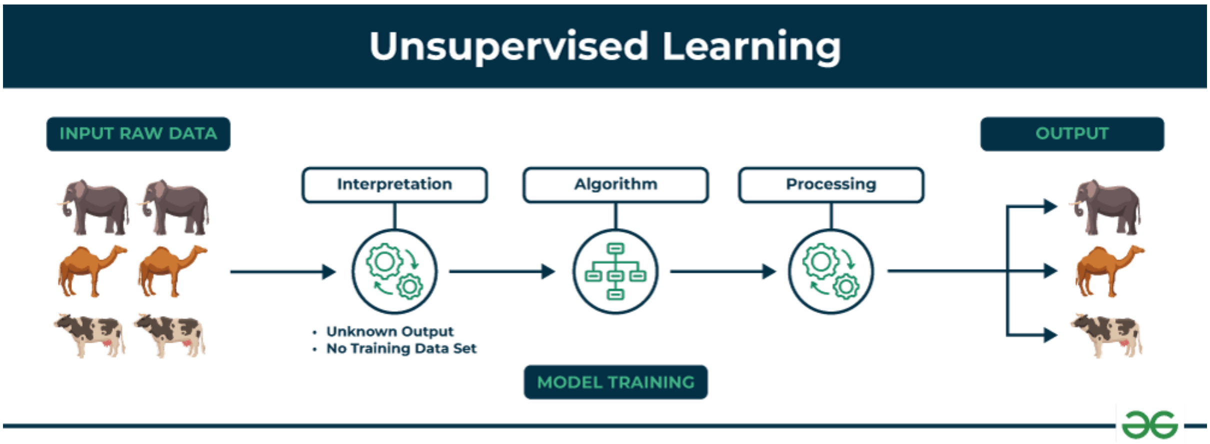
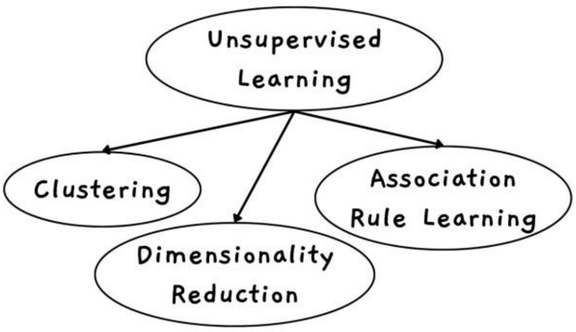


Fig. 5.1 : Unsupervised Learning

Fig. 5.1 shows unsupervised learning in which the set of animals like **elephants, camels and cows** represent raw data that the unsupervised learning algorithm will process.

Types of Unsupervised Learning



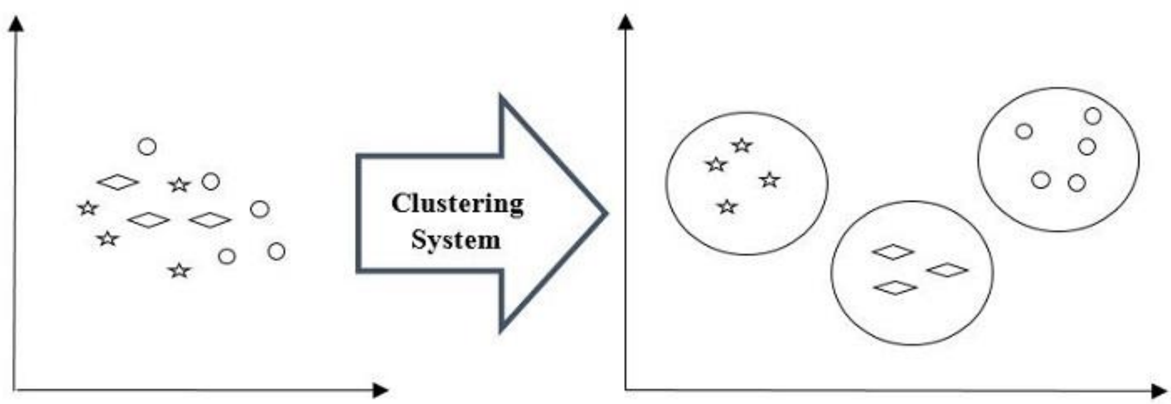
Advantages of Unsupervised Learning

1. **No Need for Labelled Data:** Works with raw, unlabeled data hence saving time and effort on data annotation.
2. **Discovers Hidden Patterns:** Finds natural groupings and structures that might be missed by humans.
3. **Handles Complex and Large Datasets:** Effective for high-dimensional or vast amounts of data.
4. **Useful for Anomaly Detection:** Can identify outliers and unusual data points without prior examples.

5.1 Clustering Techniques:-Define Clustering, Importance of clustering in data analysis, Applications of Clustering

Define Clustering

- Clustering is an **unsupervised machine learning technique that groups unlabeled data into clusters based on similarity**.
- Its goal is to **discover patterns or relationships within the data without any prior knowledge of categories or labels**.
- Clustering is the process of dividing a set of data objects into groups called clusters.
- A cluster is a group of objects that belongs to the same class. In other words, **similar objects are grouped in one cluster and dissimilar objects are grouped in another cluster**.
- The main function of clustering algorithm is to **place similar points in the same cluster while placing dissimilar points in different clusters**.
- **For example-** below is the diagram which shows clustering system grouped together the similar kind of data in different clusters −



Importance of Clustering in Data Analysis

Clustering is a technique used in **unsupervised learning** to group similar data points together. It helps in understanding large datasets by organizing data into meaningful groups.

1. Pattern Discovery

- Helps find **hidden patterns** in large data
- Groups similar data together
- Makes relationships and trends easier to understand

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2. Data Reduction

- Divides large datasets into **smaller groups (clusters)**
- Reduces complexity
- Makes data easier to analyze and visualize

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3. Preprocessing

- Used as a **first step** in machine learning
- Helps improve tasks like **classification, prediction, and analysis**

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4. Anomaly Detection

- Identifies **unusual or abnormal data points**
- Points far from clusters are considered **outliers**
- Useful in fraud detection and error finding

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5. Decision Making

- Helps organizations make **better decisions**
- Example: grouping customers for marketing strategies
- Helps in planning and strategy building

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6. Knowledge Discovery

- Helps discover **new patterns and relationships**
- Supports **innovation and new solutions**
- Useful in research and data analysis

Applications of Clustering

Clustering is useful in many fields:

1. Customer Segmentation:

Businesses use clustering to group customers based on their behavior, preferences. This helps companies create targeted marketing strategies and recommend suitable products.

2. Image Processing and Computer Vision:

In image analysis, clustering groups pixels with similar colors or intensity. This helps in tasks like image compression, object detection, and pattern recognition.

3. Document Classification and Text Mining:

Clustering organizes large collections of text documents into topics or categories. This helps search engines and websites quickly find relevant information.

4. Healthcare and Bioinformatics:

Clustering is used to group patients with similar symptoms or diseases. In bioinformatics, it helps analyze gene data and identify disease patterns.

5. Anomaly Detection:

Clustering helps detect unusual patterns or suspicious activities. It is commonly used in fraud detection and cybersecurity.

6. Social Network Analysis:

Clustering helps identify communities or groups of people with strong connections in social networks.

7. Geographical and Spatial Analysis:

Clustering divides geographic areas based on factors like climate, population, or economic activity. This helps in planning and development.

8. Recommendation Systems:

Clustering groups users with similar interests. Online platforms use this to recommend movies, products, or songs.

9. Retail and Market Basket Analysis:

Retailers use clustering to find products that are often purchased together. This helps in improving store layout and promotions.

10. Scientific Research:

Clustering helps researchers find patterns in complex data in fields like astronomy, chemistry, and environmental science.

Clustering Techniques

There are different clustering techniques used to group data.

1. Partitioning Method:

This method divides the dataset into a fixed number of clusters (k). The algorithm then improves the clusters step by step. A common example is **K-means clustering**.

2. Hierarchical Method:

This method creates a hierarchy or tree-like structure of clusters. It can either merge smaller clusters into bigger ones or divide large clusters into smaller ones.

3. Density Based Method:

This method forms clusters based on the density of data points. Areas with many data points form clusters, while areas with fewer points are considered noise. An example is **DBSCAN**.

4. Grid Based Method:

This method divides the data space into a grid of cells. Clusters are formed by combining neighboring cells that contain many data points.

5.2 K-Means Clustering: Definition and working principle, Steps involved in the K-Means algorithm, Advantages of K-Means, Disadvantages of K-Means, Hierarchical Clustering: Definition and types, Steps in Hierarchical Clustering, Advantages of Hierarchical Clustering, Disadvantages of Hierarchical Clustering, Comparing K-Means and Hierarchical Clustering

K-Means Clustering

Definition

- **K-Means clustering** is a method used to divide data into a fixed number of groups called **clusters**.
- Each cluster is represented by a **centroid**, which is the **average (mean) of all data points in that cluster**.
- The main goal of K-Means is to **group similar data points together** and **reduce the difference within each cluster**.

Operating/Working Principle

- The **K-Means algorithm** divides data into **K clusters** based on similarity.
- First, the algorithm **randomly selects K centroids**, which act as the centers of the clusters.
- Each data point is then assigned to the **nearest centroid**, usually using **Euclidean distance**.
- After assigning all points, the algorithm **recalculates the centroid** of each cluster by finding the **mean of all data points in that cluster**.
- This process **repeats multiple times**, where data points may move to different clusters and centroids are updated.
- The process stops when **centroids stop changing or cluster assignments remain the same**.

Steps Involved in the K-Means Algorithm

- K-Means is an **unsupervised machine learning algorithm** used to group **unlabeled data** into **K clusters**.
- It divides **n data points into K groups**, where each point belongs to the cluster whose **centroid is closest** to it.

Characteristics of K-Means

1. Number of Clusters

- The number of clusters **K must be chosen before running the algorithm**. Choosing the correct value of **K is important** because it affects the final clustering result.

2. Centroids

- Each cluster has a **centroid**, which represents the **average value of all points in that cluster**.

3. Iterative Process

- K-Means works in an **iterative manner**. It repeatedly **assigns data points to clusters and updates centroids** until the result becomes stable.

Algorithm Steps

Step 1: Initialize Centroids

Randomly choose **K initial centroids**.

Step 2: Assign Clusters

Calculate the distance between each data point and the centroids.

Assign each data point to the **nearest centroid**.

Step 3: Update Centroids

Recalculate the centroid of each cluster by finding the **average of all data points in that cluster**.

Step 4: Repeat

Repeat the **cluster assignment and centroid update steps until the centroids stop changing.**

Working of K-Means Clustering

- Suppose we have a **dataset containing several items with different features**.
- The objective is to **divide these items into groups based on similarity**.
- To achieve this, we use the **K-Means clustering algorithm**.
- Here **K represents the number of clusters** we want to create.

Working Process

1. Initialization- Select **K random centroids** which act as the centers of clusters.

2. Assignment Step - Each data point is assigned to the **nearest centroid**, forming clusters.

3. Update Step - For each cluster, calculate the **new centroid by taking the average of all data points** in that cluster.

4. Repeat - Repeat the assignment and update steps until the **centroids stop changing**

The goal of K-Means is to **divide the dataset into K clusters** such that:

- Data points **within the same cluster are similar**
- Data points **in different clusters are different**

Advantages and Disadvantages of K-Means Algorithm

Advantages of K-Means

1. **Simple** - K-Means is easy to understand and simple to implement.
2. **Fast** - It works faster compared to many other clustering algorithms, especially for large datasets.
3. **Efficient** - K-Means is computationally efficient and performs well when working with large amounts of data.
4. **Easy to Understand** - The idea of grouping data around **centroids (center points)** is clear and straightforward.

Disadvantages of K-Means

1. **Requires Number of Clusters** - The value of **K (number of clusters)** must be decided before running the algorithm.
2. **Sensitive to Initial Centroids**- The final clustering result depends on the **initial selection of centroids**, which may lead to different results.
3. **Difficult to Choose K**- It can be difficult to determine the **best number of clusters** for a dataset.
4. **Output Depends on Initial Input-** The final output may change depending on the **initial cluster centers chosen**.

Example of K-Means Clustering (Height and Weight)

Suppose we have data of **5 persons** based on:

- **Height (cm)**
- **Weight (kg)**

Person	Height (cm)	Weight (kg)
P1	150	50
P2	155	55
P3	160	60
P4	180	80
P5	175	75

Number of clusters

K = 2

Step 1: Initialize Centroids

Choose two initial centroids.

C1 = **(150, 50)**

C2 = **(180, 80)**

Step 2: Calculate Euclidean Distance

Formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Distance from C1 (150,50)

Person	Calculation	Distance
P1	$\sqrt{(150-150)^2+(50-50)^2}$	0
P2	$\sqrt{(155-150)^2+(55-50)^2}$	7.07
P3	$\sqrt{(160-150)^2+(60-50)^2}$	14.14
P4	$\sqrt{(180-150)^2+(80-50)^2}$	42.43
P5	$\sqrt{(175-150)^2+(75-50)^2}$	35.35

Distance from C2 (180,80)

Person	Calculation	Distance
P1	$\sqrt{(150-180)^2+(50-80)^2}$	42.43
P2	$\sqrt{(155-180)^2+(55-80)^2}$	35.35
P3	$\sqrt{(160-180)^2+(60-80)^2}$	28.28
P4	$\sqrt{(180-180)^2+(80-80)^2}$	0
P5	$\sqrt{(175-180)^2+(75-80)^2}$	7.07

Step 3: Assign Clusters

Person	Distance C1	Distance C2	Cluster
P1	0	42.43	C1
P2	7.07	35.35	C1
P3	14.14	28.28	C1
P4	42.43	0	C2
P5	35.35	7.07	C2

Clusters formed:

Cluster 1 → **P1, P2, P3**

Cluster 2 → **P4, P5**

Step 4: Update Centroids (Mean Calculation)

Cluster 1

Points:

(150,50)

(155,55)

(160,60)

Mean calculation:

$$Height = (150 + 155 + 160)/3 = 155$$
$$Weight = (50 + 55 + 60)/3 = 55$$

New centroid

C1 = (155 , 55)

Cluster 2

Points:

(180,80)

(175,75)

Mean calculation:

$$Height = (180 + 175)/2 = 177.5$$
$$Weight = (80 + 75)/2 = 77.5$$

New centroid

C2 = (177.5 , 77.5)

Final Clusters

Cluster 1 → **P1, P2, P3** (Shorter & lighter persons)

Cluster 2 → **P4, P5** (Taller & heavier persons)

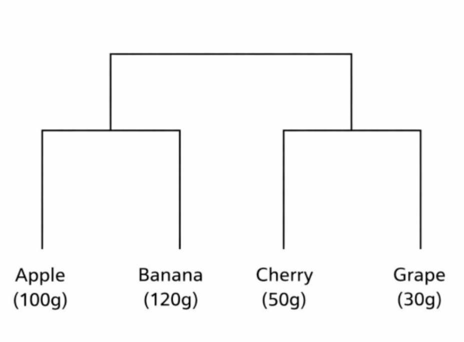
Hierarchical Clustering

- **Hierarchical Clustering** is an **unsupervised learning technique** that groups similar data points into clusters and forms a **hierarchy of clusters**.
- It creates a **tree-like structure called a dendrogram**, which shows how clusters are formed step by step.
- It **does not require selecting the number of clusters in advance**.
- It uses **two approaches: Agglomerative and Divisive**.
- It is used for **data exploration and finding hidden patterns**.
- Common applications include **pattern recognition, customer segmentation, and image grouping**.

Example - Suppose we have four fruits with different weights:

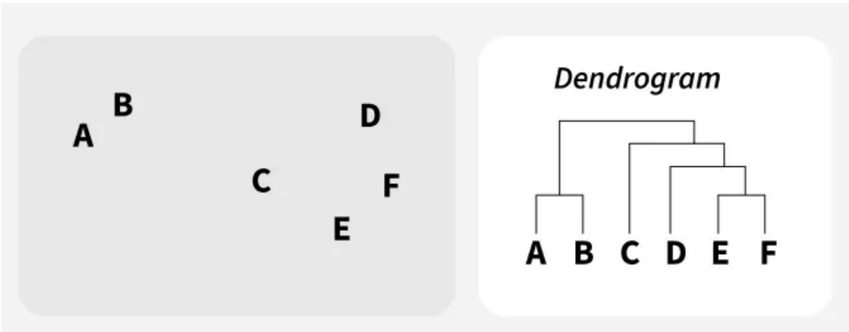
Fruit	Weight
Apple	100 g
Banana	120 g
Cherry	50 g
Grape	30 g

- In hierarchical clustering, each fruit is first treated as its own cluster.
- First, the closest fruits grape (30g) and cherry (50g) are grouped together.
- Next, apple (100g) and banana (120g) are grouped together.
- Finally, these two clusters are merged into one large cluster.
- This shows how hierarchical clustering gradually groups the most similar data points together.



Dendrogram

- A **dendrogram** is a tree-like diagram used in hierarchical clustering.
- It shows **how data points or clusters are joined together step by step based on similarity**.
- The **bottom** shows individual data points, and as we move **upwards**, similar points are merged into clusters.



In the diagram, we have **six data points: A, B, C, D, E, F**.

- **A and B merge first** because they are the most similar.
- **E and F merge together** because they are close to each other.
- **D joins the cluster (E, F)**.
- **C joins that cluster**.
- Finally, **all clusters merge into one large cluster**.

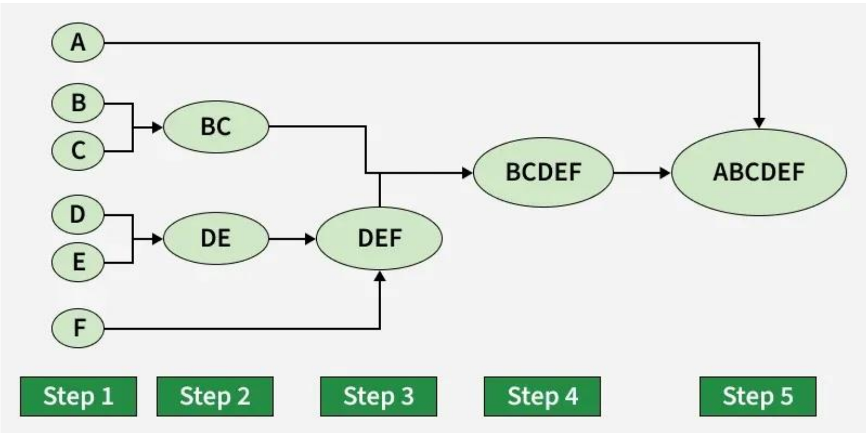
Types of Hierarchical Clustering

There are **two main types of hierarchical clustering**:

1. **Agglomerative Clustering**
2. **Divisive Clustering**

1. Hierarchical Agglomerative Clustering

- It is also called the **bottom-up approach** or **Hierarchical Agglomerative Clustering (HAC)**.
- In this method, **each data point starts as a single cluster**. Then the **closest clusters are merged step by step** until all data points are combined into **one large cluster**.
- The diagram simply shows how clusters merge step by step in Agglomerative Hierarchical Clustering.



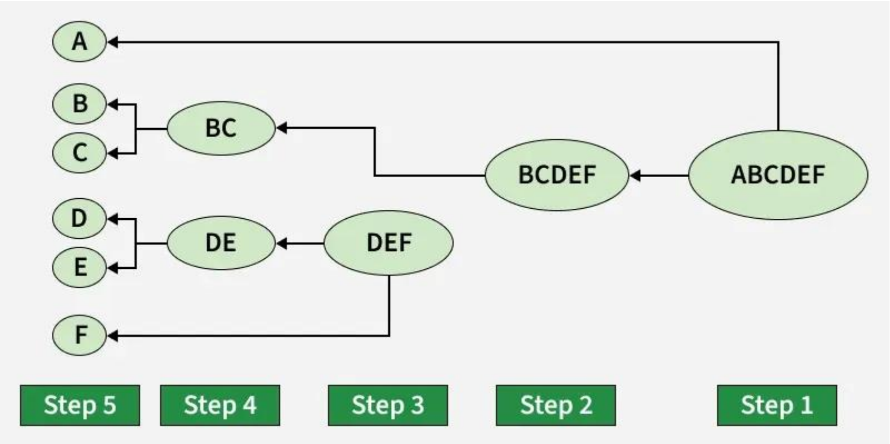
Workflow of Hierarchical Agglomerative Clustering

1. **Start with individual points**
   Each data point is treated as its **own cluster**.
   For example, if we have **5 data points**, we start with **5 clusters**.
2. **Calculate distances between clusters**
   Calculate the **distance between every pair of clusters**.
3. **Merge the closest clusters**
   Find the **two clusters with the smallest distance** and merge them into one cluster.
4. **Update distance matrix**
   After merging, **recalculate the distance** between the new cluster and the remaining clusters.

5. **Repeat the process**
   Continue merging the **closest clusters** and updating distances until **only one cluster remains**.
6. **Create a dendrogram**
   The clustering process can be shown using a **dendrogram**, which is a **tree-like diagram** showing how clusters are merged step by step.

2. Hierarchical Divisive Clustering

- Divisive clustering is also called the **top-down approach**.
- In this method, **all data points start in one large cluster**.
- Then the cluster is **split step by step into smaller clusters** until each data point becomes its own cluster.
- The diagram shows how one large cluster is split step by step into smaller clusters in Divisive Hierarchical Clustering (top-down approach).



Workflow of Hierarchical Divisive Clustering

1. **Start with one cluster**
   Treat the **entire dataset as one large cluster**.
2. **Split the cluster**
   Divide the cluster into **two smaller clusters** based on dissimilarity.
3. **Repeat the splitting**
   Select a cluster and **split it again into smaller clusters**.
4. **Continue the process**
   Keep dividing clusters **recursively**.
5. **Stop condition**
   The process stops when **each data point becomes its own cluster** or when the **required number of clusters is reached**.

Working of Hierarchical Clustering

- Hierarchical clustering is an unsupervised learning algorithm that groups data into a hierarchy of clusters. It does not require a fixed number of clusters in advance**.**

Working / Approach

Hierarchical clustering can be done in two ways:

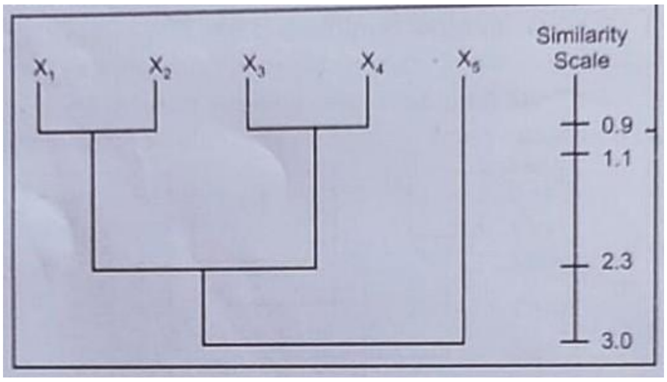
1. **Agglomerative (Bottom-Up)**
   - Starts with each data point as its **own cluster**.
   - **Smaller clusters are merged** step by step to form larger clusters.

2. **Divisive (Top-Down)**
   - Starts with **all data points in a single large cluster**.
   - **The large cluster is split** into smaller clusters recursively.

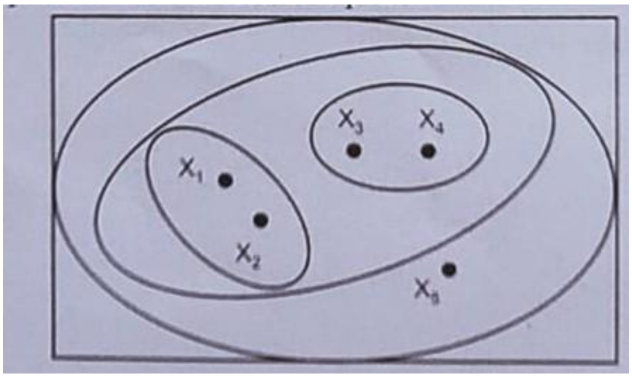
Cluster Hierarchy & Representation

- Clusters are arranged in a **hierarchy**, showing how smaller clusters combine or how larger clusters split.
- The hierarchy can be visualized using:

1. **Dendrogram**
   - A **tree-like diagram** showing clusters and their relationships.
   - The **height/length** in the dendrogram represents the **distance or similarity** between clusters.
   - Cutting the dendrogram at different levels shows **different cluster groupings**.



2. **Venn Diagram (Alternative Simple Representation)**
   - Circles are drawn around data points belonging to the **same cluster**.
   - Useful for a **simple visual understanding** of clusters.



Similarity Measure: Hierarchical clustering uses a **distance or similarity scale** to group clusters.

- **Algorithms:**
    1. **Agglomerative** – merges smaller clusters (bottom-up).
    2. **Divisive** – splits larger clusters (top-down).
- **Result:** A hierarchy that helps **understand relationships and structure** in the data.

Advantages

1. **No Need to Specify Number of Clusters**
   In hierarchical clustering, we do not need to decide the number of clusters beforehand.
2. **Visual Representation**
   The **dendrogram** gives a clear visual picture of how clusters are formed and related.
3. **Easy to Understand Hierarchy**
   The dendrogram shows a **tree-like structure**, making it easy to understand the relationship between clusters at different levels.

Disadvantages

1. **Computationally Expensive**
   Hierarchical clustering requires a lot of computation, especially for **large datasets**.
2. **Less Flexible**
   Once clusters are formed, the structure **cannot be changed easily**.
3. **Sensitive to Distance and Linkage Method**
   The result depends heavily on the **distance measure and linkage method** used.

4. **Difficult with Large Datasets**
   When the dataset is very large, the **dendrogram becomes complex and hard to interpret**.
5. **Cannot Undo Steps**
   Once two clusters are merged or split, the algorithm **cannot reverse that step**.

Difference between K-Means Clustering and Hierarchical Clustering

Feature	K-Means Clustering	Hierarchical Clustering
Definition	A method that **divides data into K groups** based on similarity	A method that **builds a hierarchy (tree) of clusters** to show relationships
Approach	**Partitioning:** starts with K centroids and updates them iteratively	**Hierarchical:** merges (agglomerative) or splits (divisive) clusters step by step
Clusters form	Starts with K points (centroids) and updates	Step-by-step merging or splitting
Number of clusters	Must decide K before starting	Can choose later by cutting the dendrogram
Cluster shape	Best for round/equal clusters	Can handle irregular or nested clusters
Speed & size	Fast, works for large data	Slow, better for small/medium data
Stability	Depends on starting points	More stable, but affected by outliers
Interpretation	Only final clusters	Shows full hierarchy of clusters
Memory	Low	High
Best for	Large data with known groups	Understanding structure and relationships in data
Example	Grouping students by marks	Organizing animals by type

5.3 Dimensionality Reduction: Importance of Dimensionality Reduction

Dimensionality Reduction

- Dimensionality reduction is an important concept in machine learning used to simplify complex datasets.
- It is the process of **reducing the number of input variables (features)** in a dataset while **preserving the most important information**. By reducing the number of features, the data becomes easier to analyze, visualize, and process.
- The main goal of dimensionality reduction is to **simplify the dataset.**
- This helps, the data becomes easier to analyse, visualize, and process. Also reduce computational cost, improve model performance

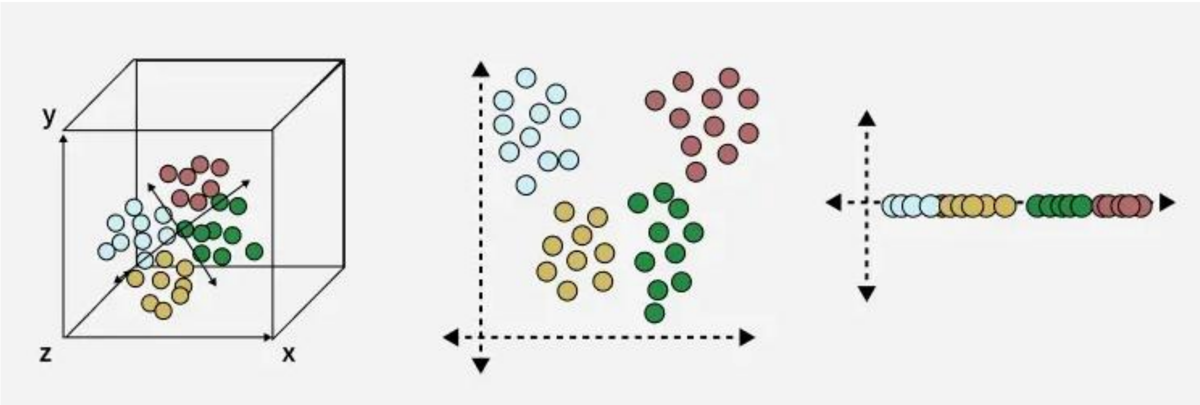
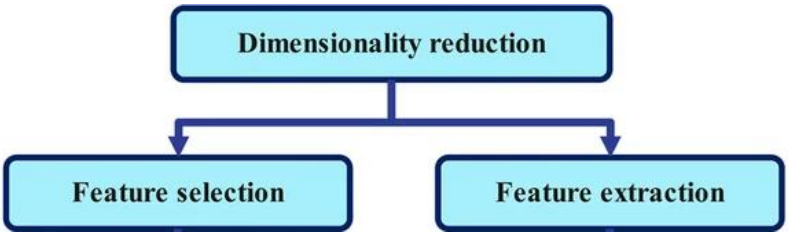


Fig.- Dimensionality reduction converts **high-dimensional data into lower dimensions (3D → 2D → 1D)** while preserving important information.

Two Main Types/Approaches to Dimensionality Reduction



1. Feature Selection

- Feature selection is the process of **choosing the most important features from the dataset**.
- It selects useful features **without changing the original data**.
- It helps remove **irrelevant or duplicate features**, making the model faster and more efficient.

Common Methods

1. **Filter Method:** Selects features based on their importance or correlation with the target variable.
2. **Wrapper Method:** Selects features based on the **performance of the machine learning model**.
3. **Embedded Method:** Feature selection happens **during the model training process**.

2. Feature Extraction

- Feature extraction is the process of **creating new features from the original features**.
- It **combines or transforms existing features** to reduce the number of variables.
- The new features **retain most of the important information** in fewer dimensions.

Common Methods

1. **Principal Component Analysis (PCA):** Converts correlated variables into new independent variables called **principal components**.
2. **Backward Feature Elimination:** Starts with all features and removes the **least important features step by step**.
3. **Forward Feature Selection:** Starts with one feature and **adds more features gradually** based on performance.
4. **Random Forest:** Uses decision trees to **identify and select the most important features automatically**.

- Dimensionality reduction is mainly required when datasets contain a **large number of features**, which can increase model complexity, slow down training, and lead to problems like **overfitting**.
- In simple terms, dimensionality reduction **transforms high-dimensional data into a lower-dimensional representation without losing important information**.

Importance of Dimensionality Reduction

1. Reduces Computational Complexity

High-dimensional data needs more memory and processing time. Reducing the number of features makes algorithms faster and easier to run.

2. Removes Redundant and Irrelevant Data

Some features may be repeated or not useful. Removing them improves the quality of the dataset and reduces overfitting.

3. Improves Model Performance

With fewer and more important features, the model becomes simpler and can make better predictions.

4. Helps in Visualization

When data has many features, it is hard to visualize. Dimensionality reduction converts it into **2D or 3D**, making patterns and clusters easier to see.

5. Reduces Data Storage (Compression)

Fewer features mean less storage is required, which is helpful for large datasets like images, videos, and text.

6. Improves Data Understanding (Interpretability)

Lower-dimensional data is easier to analyze and helps identify the most important features.

7. Solves the Curse of Dimensionality

When the number of features is very high, data becomes sparse and difficult to analyze. Reducing dimensions makes the data more meaningful and manageable.

5.4 Dimensionality Reduction 5.4 PCA -Definition and fundamental principles of PCA, Eigenvectors and eigenvalues, Steps in PCA, Explained Variance, Choosing the Optimal Dimensionality, Advantages and Disadvantages of PCA, Applications of PCA

PCA (Principal Component Analysis)

- **Principal Component Analysis (PCA)** is a technique used in machine learning to **reduce the number of features in a dataset while keeping most of the important information**.
- It converts the original features into **new features called principal components**.

Objectives of PCA

1. **Reduce Dimensionality:** Reduce the number of features while keeping important information.
2. **Eliminate Redundancy:** Remove similar or highly related features.
3. **Improve Efficiency:** Fewer features make machine learning algorithms faster.
4. **Help in Visualization:** Convert high-dimensional data into 2D or 3D for easier visualization.

Fundamental Principles of PCA

1. **Dimensionality Reduction**
   PCA reduces the **number of features** in the dataset.
2. **Principal Components**
   It creates new variables called **principal components (PC1, PC2, etc.)**.
3. **Maximum Variance**
   The first principal component contains the **maximum information (variance)**.
4. **Uncorrelated Components**
   The principal components are **independent from each other**.
5. **Data Transformation**
   PCA transforms the original data into a **new coordinate system**.
6. **Data Standardization**
   Before applying PCA, data is **standardized so all features have equal importance**.

Eigenvectors and Eigenvalues

In machine learning, **eigenvalues and eigenvectors** are used to understand data patterns.
They are mainly used in **dimensionality reduction techniques like PCA**.

- **Eigenvectors** show the **important direction in the data**.
- **Eigenvalues** show **how important that direction is**.

These concepts come from **linear algebra** and are used in **machine learning and artificial intelligence**.

They are used in applications such as:

- Image recognition
- Natural language processing
- Robotics

Mathematical form:

$$Av = \lambda v$$

This means when matrix **A** is multiplied by vector **v**, the result is **λ times v**.

Mathematical Calculation

To find **eigenvalues**:

$$|A - \lambda I| = 0$$

To find **eigenvectors**:

$$(A - \lambda I)v = 0$$

Where:

A = matrix

λ = eigenvalue

I = identity matrix

v = eigenvector

Uses of Eigenvalues and Eigenvectors

1. **Dimensionality Reduction (PCA)** – Reduce the number of features while keeping important information.
2. **Image Compression** – Reduce the size of images for easier storage.
3. **Support Vector Machines (SVM)** – Improve classification in machine learning models.
4. **Graph Analysis** – Study networks and find important nodes.
5. **Natural Language Processing (NLP)** – Identify important words and patterns in large text data.

Steps in PCA

1. Standardize the Data

All features are converted to the **same scale**
(mean = 0 and standard deviation = 1).
Example: Height (150–180) and Weight (50–70) are normalized.

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2. Calculate Covariance Matrix

This step finds the **relationship between features**.
It shows:

- Which features are related
- How strongly they are related

Example: Height and Weight are positively related.

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3. Find Eigenvalues and Eigenvectors

From the covariance matrix, calculate:

- **Eigenvectors → Direction of data**
- **Eigenvalues → Importance of that direction**

This step identifies the **main directions of data spread**.

---

4. Select Top Principal Components

Select eigenvectors with the **highest eigenvalues**.
Example:
If PC1 = 50% and PC2 = 30%,
then PC1 + PC2 = **80% information**, so select them.

---

5. Transform the Data

Convert original data into **new features called Principal Components**.
Result:

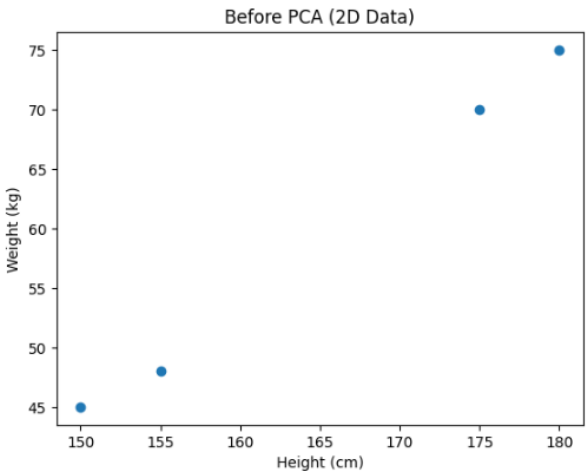
- Reduced number of features
- Important information is preserved

Example- Features- Height, Weight
Both features give similar information about body size.

Height (cm)	**Weight (kg)**
150	45
155	48
175	70
180	75

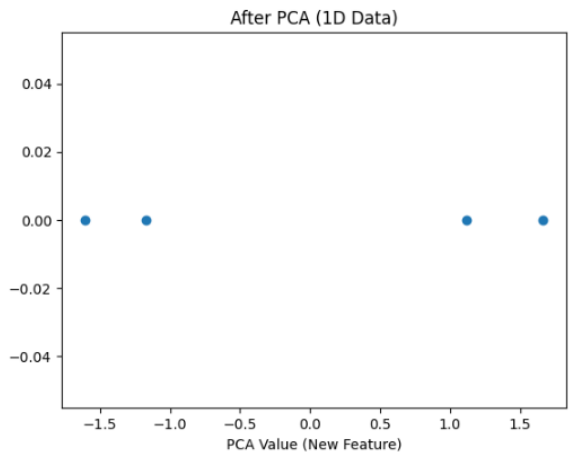
Before PCA Graph - Height and Weight increase together.

- Two axes → Height & Weight
- Data is 2D
- Points spread diagonally
- Both features give similar information“



After PCA Graph - PCA combined two features into one

1. Only **one axis**
2. PCA **combined Height & Weight**
3. New feature represents **overall body size**



Explained Variance

- Explained variance tells how much information each principal component (PC) contains.
- In PCA, original data is converted into principal components (PCs).
- Each PC contains some part of the total information (variance).
- The eigenvalue tells how much information a PC has.
- Bigger eigenvalue means more important component
- So, PCs with higher explained variance are considered more important.

Example

Principal Component	Explained Variance
PC1	50%
PC2	30%
PC3	10%
PC4	7%
PC5	3%

Here:

- PC1 explains **50% of the variance**
- PC2 explains **30% of the variance**

Together **PC1 + PC2 explain 80% of the total information** in the dataset.

Choosing the Optimal Dimensionality

- Choosing optimal dimensionality means **selecting the right number** of principal components
- So that we keep most of the important information and reduce the number of features
- The aim is to **reduce the dataset size without losing important information**.
- If too many components are kept, dimensionality reduction is not effective.
- If too few components are selected, important information may be lost.

How to Choose Optimal Components

1. Explained Variance Method

Add principal components until **80%–95% variance** is covered

Example:

PC1 = 50%

PC2 = 30%

PC3 = 10%

PC1 + PC2 = **80%**

So we can keep **2 principal components instead of all features**.

---

2. Scree Plot Method

- A **Scree Plot** is a graph that shows the explained variance for each principal component.
- Choose components where the graph starts to changing very little.

Advantages and Disadvantages of PCA

Advantages of PCA

1. **Reduces Features (Dimensionality)**
   PCA reduces the number of features but keeps important information.

2. **Removes Similar Data (Correlated Features)**
   It removes correlated (similar) features.

3. **Easy to Understand Data**
   With fewer features, data becomes easier to understand.

4. **Reduces Overfitting**
   Helps the model perform better on new data.

5. **Faster Computation**
   Fewer features reduce computation time, making training faster.

6. **Handles Multicollinearity**
   Converts correlated features into independent ones.

7. **Removes Noise**
   Removes less important (low variance) data.

8. **Data Compression**
   Reduces storage space and speeds up processing.

9. **Detects Outliers**
   Helps find unusual data points.

10. **Better Visualization**
    Converts data into **2D or 3D** for easy visualization.

11. **Improves Model Performance**
    Improves efficiency and sometimes accuracy.

Disadvantages of PCA

1. **Hard to Understand**
   New components are difficult to interpret.

2. **Needs Data Scaling**
   Data must be normalized before applying PCA.

3. **Loss of Information**
   Some important data may be lost.

4. **Works on Linear Data**
   Not suitable for complex (non-linear) data.

5. **High Computation for Large Data**
   Can be slow for very large datasets.

6. **Difficult Interpretation**
   New features are combinations, not original ones.

7. **Assumes Linear Relationship**
   Works only when features are linearly related.

8. **Requires Standardization**
   Data must be properly scaled.

Applications of PCA

1. **Data Visualization**
   Converts high-dimensional data into **2D/3D** for easy understanding.

2. **Image Compression**
   Reduces image size with minimum quality loss.

3. **Noise Removal**
   Removes unwanted noise from signals like audio.

4. **Feature Extraction**
   Used before training ML models to reduce features.

5. **Bioinformatics**
   Used to analyze genes and DNA patterns.

6. **Finance**
   Helps in risk analysis and understanding market factors.

7. **Face Recognition**
   Used in systems to detect and recognize faces.

8. **Medical Analysis**
   Helps analyze medical data and find relationships.

Assignment No 5

1. Explain Unsupervised learning with example
2. Define term clustering. Give importance of clustering in data analysis.
3. Write applications of clustering.
4. Explain K-mean clustering with working and example.
5. Describe hierarchical clustering with its types. State its advantages and disadvantages.
6. What is dimensionality reduction? Give its importance.
7. Compare K-mean clustering and hierarchical clustering.
8. What is PCA? Give principles of PCA.
9. Explain importance of dimensionality reduction in detail.
10. Define term eigenvectors and eigenvalues.
11. Explain steps in PCA in detail.
12. What is explained variance.
13. Write note on Choosing the Optimal Dimensionality
14. Write advantages and disadvantages of PCA.
15. Give applications of PCA.